

DESIGN AND IMPLEMENTATION OF A FEDERATED LEARNING MODEL FOR INTRUSION DETECTION SYSTEMS BASED DATA GENERATION

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ABSTRACT

The increasing complexity and frequency of cyber-attacks have necessitated the development of more effective and scalable intrusion detection systems (IDS). Traditional centralized IDS approaches face significant challenges, including data privacy concerns, high communication overhead, and poor scalability in distributed environments. In addition, the presence of imbalanced datasets in intrusion detection tasks often leads to biased models that fail to accurately detect minority attack classes.

This study presents the design and implementation of a federated learning-based intrusion detection system that incorporates data generation techniques to address these challenges. The proposed system utilizes a federated learning framework, where multiple client nodes independently train local models using their respective datasets, while a central server aggregates the model parameters to form a global model. To improve the representation of minority classes, the Synthetic Minority Over-sampling Technique (SMOTE) is applied at the client level before model training.

The NSL-KDD dataset is used for training and evaluation, with the intrusion detection task formulated as a binary classification problem. The performance of the proposed system is evaluated using standard metrics, including accuracy, precision, recall, and F1-score. A comparative analysis is conducted between models trained with and without SMOTE to assess the impact of data generation on detection performance.

The study concludes that combining federated learning with efficient data generation techniques provides a practical and privacy-preserving solution for intrusion detection in distributed environments.

CHAPTER ONE

INTRODUCTION

1.1 Background of the study

The increasing reliance on computer networks, cloud computing, and Internet of Things (IoT) systems has led to a corresponding rise in cyber threats, making network security a critical concern. Intrusion Detection Systems (IDS) are widely used to monitor network traffic and identify malicious activities such as unauthorized access and denial-of-service attacks (Azeez et al., 2019). Traditional IDS approaches typically rely on centralized architectures, where data from multiple sources is collected and analyzed in a single location. However, such systems face challenges related to data privacy, high communication overhead, and limited scalability in distributed environments.

To enhance detection capabilities, machine learning techniques have been integrated into IDS, enabling systems to automatically learn patterns from network traffic data. Despite their effectiveness, these approaches often require large, well-balanced datasets. In practice, intrusion detection datasets are highly imbalanced, with significantly fewer attack instances compared to normal traffic, which can lead to biased models and reduced detection accuracy for minority classes (Bhowmick & Hazarika, 2016). Additionally, the distributed nature of modern network environments makes centralized data collection impractical due to privacy and regulatory constraints.

Federated Learning (FL) has emerged as a promising solution to these challenges by enabling multiple clients to collaboratively train a global model without sharing raw data. Instead, each client trains a local model using its own data and shares only model updates with a central server for aggregation, thereby preserving privacy and reducing communication costs (Abhijit et al., 2025). This makes federated learning particularly suitable for intrusion detection in decentralized environments such as IoT networks.

However, federated learning introduces challenges such as data heterogeneity (non-IID data) and class imbalance across clients, which can negatively affect model performance. To address the issue of imbalanced data, data generation techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) are employed. SMOTE generates synthetic samples for minority classes by interpolating between existing data points, thereby improving class distribution and enhancing model learning (Chawla et al., 2002).

1.1.1 more information

In this study, a federated learning-based intrusion detection model is developed, incorporating SMOTE as a data generation technique at the client level to address class imbalance. The system enables distributed clients to perform local data augmentation and model training, followed by global model aggregation. By integrating federated learning with SMOTE-based data generation, the study aims to improve intrusion detection performance while maintaining data privacy and scalability in dis-

tributed environments.

1.2 Statement of Problem

The increasing sophistication and frequency of cyber-attacks have made effective intrusion detection a major concern in modern networked systems. While Intrusion Detection Systems (IDS) play a vital role in identifying malicious activities, traditional IDS approaches based on centralized architectures face significant limitations. These systems require the aggregation of large volumes of network data at a central location, which raises serious concerns regarding data privacy, communication overhead, and scalability, especially in distributed environments such as cloud and IoT networks (Azeez et al., 2019).

Although machine learning-based IDS models have improved detection capabilities, their effectiveness largely depends on the availability of high-quality and balanced datasets. In practice, intrusion detection datasets are often highly imbalanced, with significantly fewer instances of attack traffic compared to normal traffic. This imbalance leads to biased learning, where models tend to favor the majority class, resulting in poor detection of minority attack types and increased false negatives (Bhowmick & Hazarika, 2016). Consequently, many IDS models fail to effectively detect rare but critical intrusion patterns.

Furthermore, in real-world distributed systems, data is naturally decentralized across multiple nodes or devices, making centralized training impractical or undesirable due to privacy and regulatory constraints. Fed-

erated Learning (FL) has been introduced to address these issues by enabling collaborative model training without sharing raw data. However, federated learning itself is affected by challenges such as non-independent and identically distributed (non-IID) data across clients and uneven class distributions, which can degrade the performance of the global model (Abhijit et al., 2025).

Despite existing efforts to improve federated intrusion detection systems, there remains a need for effective mechanisms to handle data imbalance within the federated learning framework. Data generation techniques have been proposed to address this issue by augmenting minority class samples, thereby improving model performance. However, the integration of such techniques within federated environments is still an area requiring further exploration, particularly with respect to their effectiveness and practicality in distributed settings.

Therefore, the key problem addressed in this study is the development of a federated learning-based intrusion detection model that can effectively handle data imbalance across distributed clients while maintaining data privacy and improving overall detection performance. This study investigates the use of SMOTE-based data generation within a federated learning framework to enhance the detection of minority attack classes and improve the robustness of intrusion detection systems.

1.3 Aim and Objectives

The aim of this study is to design and implement a federated learning model for intrusion detection systems that incorporates data generation techniques to improve detection accuracy, address data imbalance, and enhance performance in distributed environments.

Objectives

The formulated objectives are to:

- i. Incorporate SMOTE-based data generation into a federated learning framework to improve the representation of minority attack classes during local model training.
- ii. Design and implement a federated intrusion detection system (IDS) that enables collaborative learning across multiple distributed clients while preserving data privacy.
- iii. Evaluate the performance of the proposed model using benchmark intrusion detection datasets under federated learning conditions.
- iv. Compare the proposed model with baseline approaches, including federated learning without data generation, using performance metrics such as accuracy, precision, recall, and F1-score.

1.4 Scope of the study

This study focuses on the design and implementation of a federated learning-based intrusion detection system that incorporates data generation techniques to improve detection performance in distributed environments.

The research is specifically limited to the development and evaluation of a machine learning model trained using a federated learning framework, where multiple clients independently train local models and contribute to a global model through parameter aggregation.

The study utilizes the NSL-KDD dataset as the primary benchmark for intrusion detection. The dataset is preprocessed to suit the requirements of machine learning algorithms, including feature encoding, normalization, and label transformation. For simplicity and effective evaluation, the intrusion detection task is formulated as a binary classification problem, where network traffic is categorized as either normal or attack.

The federated learning setup in this study is simulated using multiple client nodes, each representing a distributed data source. Data is partitioned across these clients to emulate real-world distributed environments, although the simulation is conducted on a single machine rather than a physically distributed network. The federated learning process involves local model training at each client and global model aggregation using a standard federated averaging (FedAvg) approach.

To address the issue of class imbalance within the dataset, the study incorporates the Synthetic Minority Over-sampling Technique (SMOTE) as a data generation method. SMOTE is applied locally at each client to generate synthetic samples for minority classes before model training. The impact of this data generation technique is evaluated by comparing model performance with and without SMOTE.

The performance of the proposed system is evaluated using standard clas-

sification metrics, including accuracy, precision, recall, and F1-score. Additionally, confusion matrices and graphical plots are used to provide a visual representation of the model's performance.

1.5 Significance of the Study

This study is significant due to the growing need for effective, scalable, and privacy-preserving intrusion detection systems in modern distributed network environments. As cyber threats continue to evolve in complexity and frequency, there is an increasing demand for intelligent security solutions capable of detecting both common and rare attack patterns. This research contributes to the advancement of intrusion detection by integrating federated learning with data generation techniques, thereby addressing key limitations in existing approaches.

Firstly, the study contributes to the field of cybersecurity by proposing a federated learning-based intrusion detection model that enhances data privacy. Unlike traditional centralized IDS approaches that require the transfer of raw network data to a central server, the proposed system allows clients to train models locally and share only model updates. This reduces the risk of data leakage and aligns with modern data protection requirements, making the approach suitable for sensitive environments such as financial systems, healthcare networks, and IoT infrastructures (Abhijit et al., 2025).

Secondly, the study addresses the critical issue of data imbalance in intrusion detection datasets. By incorporating SMOTE-based data generation

at the client level, the proposed system improves the representation of minority attack classes, thereby enhancing the model's ability to detect rare but potentially harmful intrusions. This contributes to improved classification performance, particularly in terms of recall and F1-score, which are essential for evaluating intrusion detection systems (Bhowmick & Hazarika, 2016).

Furthermore, the study provides a practical implementation framework for combining federated learning and data generation techniques in a distributed setting. The use of benchmark datasets such as NSL-KDD demonstrates the applicability of the proposed model in real-world scenarios. The experimental results obtained from the prototype offer empirical evidence of the effectiveness of integrating SMOTE within a federated learning environment, thereby serving as a reference for future research in this area.

1.6 Justification of the Study

The need for this study arises from the limitations of existing intrusion detection systems in handling privacy, scalability, and data imbalance in modern distributed network environments. Traditional centralized intrusion detection approaches require the collection of large volumes of network traffic data at a central location, which exposes sensitive information to potential breaches and increases communication overhead. With the growing adoption of distributed systems such as cloud computing and Internet of Things (IoT) networks, there is a clear need for intrusion

detection mechanisms that can operate efficiently without compromising data privacy (Azeez et al., 2019).

Federated Learning provides a suitable framework for addressing these concerns by enabling collaborative model training across multiple clients without sharing raw data. However, despite its advantages, federated learning alone does not fully solve the problem of imbalanced datasets, which is a common characteristic of intrusion detection data. In such datasets, attack instances—especially rare or sophisticated attacks are significantly underrepresented, leading to poor detection performance and increased risk of undetected intrusions (Bhowmick & Hazarika, 2016). This study is therefore justified by the need to integrate an effective data generation mechanism within the federated learning framework to enhance intrusion detection performance. The use of the Synthetic Minority Over-sampling Technique (SMOTE) provides a practical and computationally efficient solution for generating synthetic samples of minority classes at the client level. By improving class balance locally before model training, SMOTE helps to enhance the model’s ability to learn diverse attack patterns and improves overall classification performance. Furthermore, while advanced generative models such as Generative Adversarial Networks (GANs) and Conditional Variational Autoencoders (CVAEs) have been proposed in recent research, their implementation is often complex and computationally intensive. This study justifies the use of SMOTE as a lightweight and effective alternative that is more suitable for federated environments, especially in resource-constrained settings.

1.7 Methodology

This study adopts a practical and experimental research approach for the design and implementation of a federated learning-based intrusion detection system incorporating data generation techniques. The methodology involves the development of a machine learning model within a simulated federated learning environment, where multiple clients independently train local models using their respective datasets.

The NSL-KDD dataset is utilized as the benchmark dataset for this study. The dataset undergoes preprocessing steps including data cleaning, feature encoding, normalization, and label transformation. For effective analysis, the intrusion detection task is formulated as a binary classification problem, categorizing network traffic as either normal or attack.

The proposed system is implemented using a federated learning framework in which the training data is partitioned across multiple clients to simulate distributed data sources. Each client applies the Synthetic Minority Over-sampling Technique (SMOTE) locally to address class imbalance before training its model. The locally trained models are then aggregated at a central server using a federated averaging approach to produce a global model.

The performance of the system is evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. Additionally, confusion matrices and graphical representations are used to provide a comprehensive analysis of the model's performance. A comparative evaluation is also conducted between models trained with and without

SMOTE to assess the impact of data generation on intrusion detection performance.

1.8 Organisation of project chapters

This project is organized into five chapters, each focusing on a specific aspect of the study.

(a) Chapter One: Introduction

This chapter presents the background to the study, problem statement, aim and objectives, significance, scope, justification, and methodology of the research.

(b) Chapter Two: Literature Review

This chapter reviews relevant literature on intrusion detection systems, federated learning, and data generation techniques. It also examines related works and identifies research gaps addressed by this study.

(c) Chapter Three: Methodology

This chapter describes the design and implementation of the proposed system, including data preprocessing, federated learning framework, SMOTE integration, and system architecture.

(d) Chapter Four: Result and Discussion

This chapter presents the experimental results obtained from the implemented model. It includes performance evaluation, comparison of results, and analysis using metrics and visualizations.

(e) **Chapter Five:** Conclusion and Recommendation

This chapter summarizes the findings of the study, draws conclusions, and provides recommendations for future research.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

The increasing complexity of modern network systems has led to a corresponding rise in cyber threats, making effective intrusion detection essential for maintaining network security. Traditional intrusion detection systems (IDS) face challenges such as scalability, data privacy, and the ability to handle large volumes of distributed data. As a result, recent research has focused on integrating machine learning techniques and distributed learning approaches to improve detection performance.

This chapter reviews relevant literature on intrusion detection systems, federated learning, and data generation techniques used to address class imbalance in intrusion detection datasets. It also examines existing studies that combine these approaches and highlights the research gaps that this study aims to address.

2.2 Fundamental Concepts

The development of an effective intrusion detection system within a federated learning environment requires a clear understanding of several foundational concepts. These concepts form the basis upon which the proposed model is designed and implemented. This section presents key terms and ideas relevant to intrusion detection, federated learning, and data generation techniques, providing the necessary theoretical grounding for the study.

Intrusion detection systems are designed to monitor and analyze network activities in order to identify potential security breaches. With the increasing complexity of cyber threats, modern IDS have evolved from traditional rule-based systems to intelligent systems that leverage machine learning for improved detection accuracy (Ahmad et al., 2021). These systems are capable of identifying both known and unknown attack patterns, making them more adaptable in dynamic network environments.

Federated learning represents a paradigm shift from centralized machine learning to distributed model training. In this approach, multiple clients collaboratively train a global model without sharing raw data, thereby preserving privacy and reducing the risks associated with data centralization (Kairouz et al., 2021). This concept is particularly important in intrusion detection, where network data may be sensitive and distributed across multiple locations.

Another important concept in this study is data imbalance, which is a common issue in intrusion detection datasets. In most cases, normal network traffic significantly outweighs malicious traffic, leading to biased learning and poor detection of minority classes. To address this, data generation techniques are employed to improve class distribution and enhance model performance. Among these techniques, SMOTE remains a widely adopted method due to its simplicity and effectiveness in generating synthetic minority samples (Fernández et al., 2022).

2.2.1 Definition of Terms

For clarity and consistency, the following key terms are defined as used in this study:

Definition 2.2.1 (Intrusion Detection System (IDS):) *A system designed to monitor and analyze network or system activities in order to detect unauthorized access or malicious behavior.*

Definition 2.2.2 (Federated Learning (FL):) *A distributed machine learning approach in which multiple clients collaboratively train a global model without sharing their raw data.*

Definition 2.2.3 (Client:) *A node or device in a federated learning system that holds local data and performs local model training.*

Definition 2.2.4 (Global Model:) *The aggregated model obtained by combining updates from multiple client models in a federated learning framework.*

Definition 2.2.5 (Data Imbalance:) *A condition in which one class (e.g., normal traffic) significantly outnumbers other classes (e.g., attack traffic) in a dataset.*

Definition 2.2.6 (SMOTE (Synthetic Monitoring Over-Sampling):) *A data generation method used to create synthetic samples for minority classes in order to balance the dataset.*

Definition 2.2.7 (Classification Model:) *A machine learning model used to categorize input data into predefined classes, such as normal or attack.*

Definition 2.2.8 (Feature Engineering:) *The process of transforming raw data into meaningful features that improve the performance of machine learning models.*

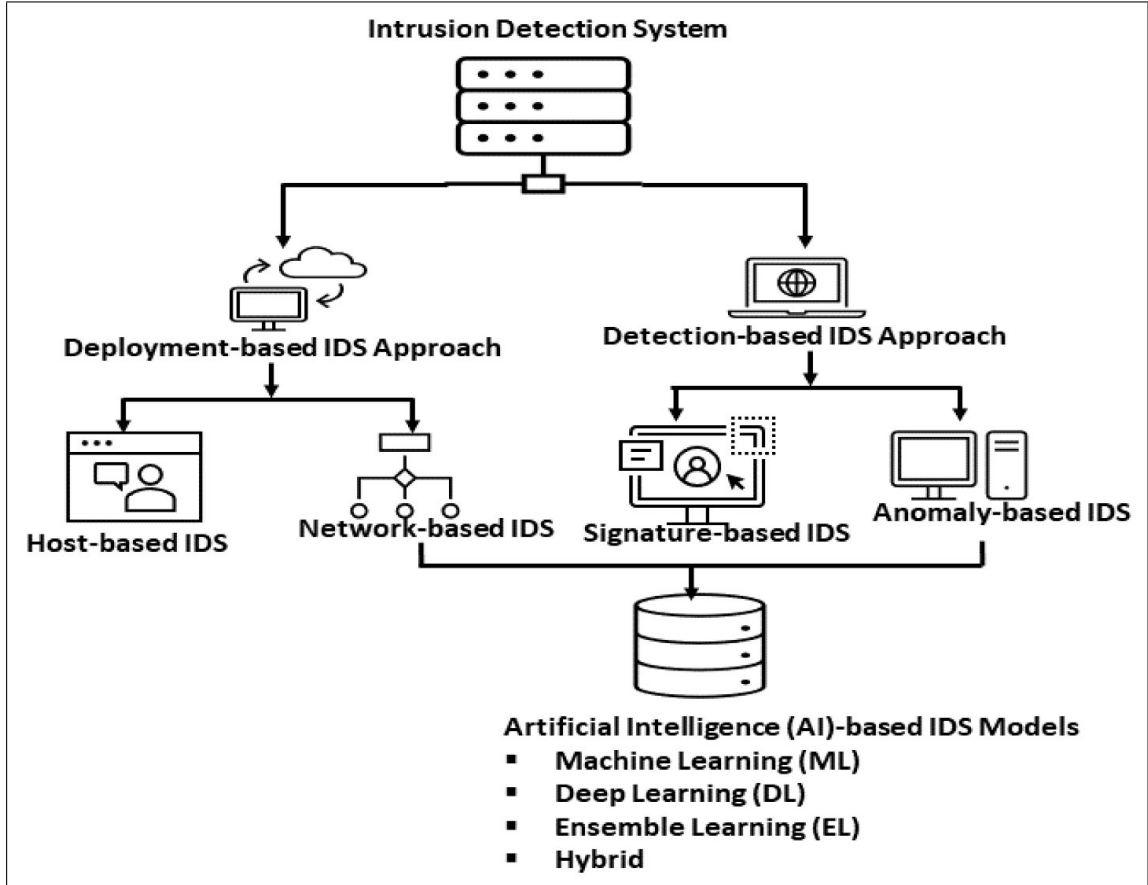


Figure 2.1: Classification of Intrusion Detection Systems

Liao, Y., Vemuri, V. R. (2013). *Use of k-nearest neighbor classifier for intrusion detection.*

(Still widely cited foundational IDS classification paper)

<https://ieeexplore.ieee.org/document/8260600>

2.3 Analysis of Theories

This section examines the key theoretical foundations that underpin the design and implementation of the proposed federated learning-based in-

trusion detection system. These theories are drawn from machine learning, distributed systems, and data imbalance handling, and they collectively justify the approach adopted in this study.

2.3.1 Supervised Machine Learning Theory

Supervised machine learning forms the basis for most modern intrusion detection systems. This theory involves training models on labeled datasets to learn the relationship between input features and corresponding output classes. In the context of intrusion detection, the model learns to classify network traffic as either normal or malicious.

The effectiveness of supervised learning depends largely on the availability of high-quality labeled data and balanced class distributions. Models trained under this paradigm aim to minimize classification errors by optimizing a predefined loss function. Recent studies have shown that supervised learning remains highly effective for intrusion detection when appropriate preprocessing and feature engineering techniques are applied (Ahmad et al., 2021).

2.3.2 Federated Learning and Distributed Optimization Theory

Federated learning is grounded in distributed optimization theory, where multiple clients collaboratively train a global model without sharing raw data. Each client performs local computations using its dataset and transmits model updates to a central server. These updates are then aggregated to form an improved global model.

The commonly used Federated Averaging (FedAvg) algorithm operates

by averaging the parameters of local models across clients to approximate a centralized learning solution. While this approach enhances data privacy and reduces communication costs, it introduces challenges such as non-independent and identically distributed (non-IID) data, which can affect model convergence and performance (Kairouz et al., 2021).

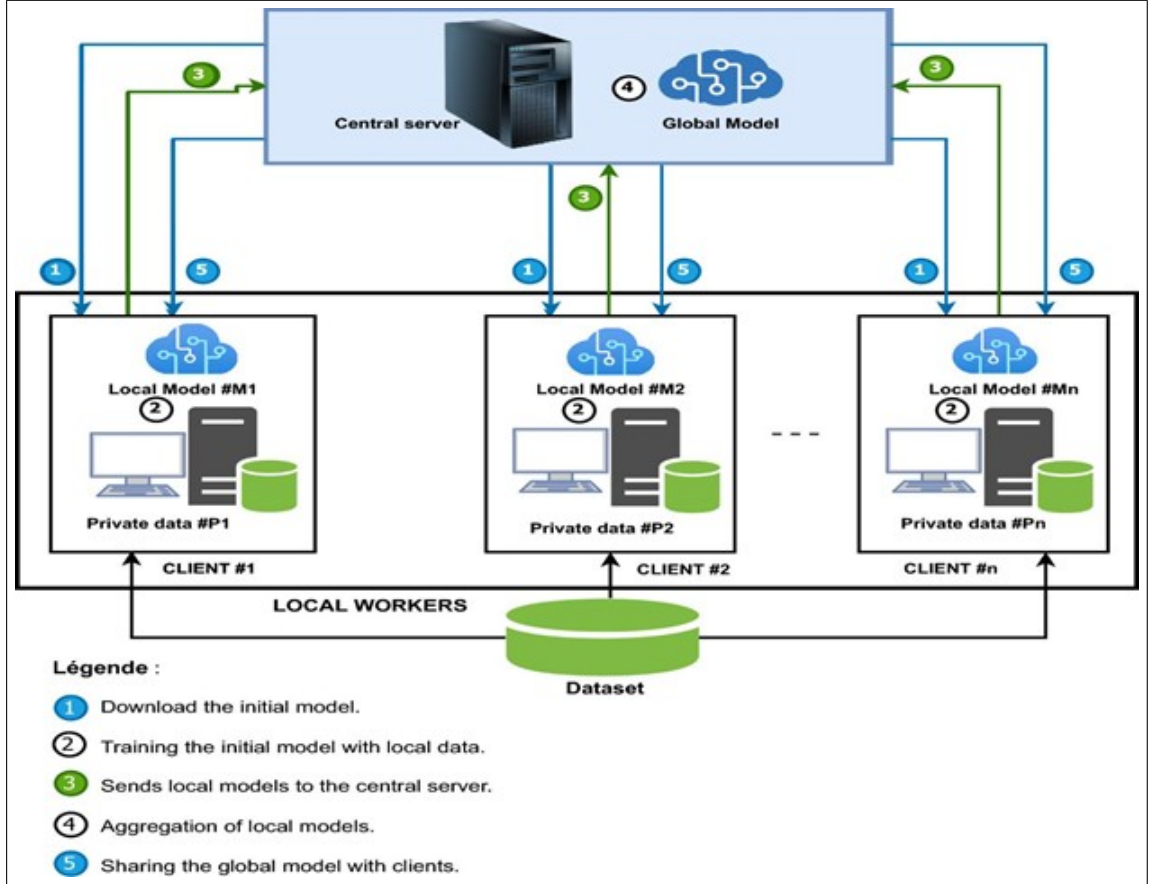


Figure 2.2: Federated Learning Architecture
<https://arxiv.org/abs/1602.05629>

2.3.3 Theory of Data Imbalance in Machine Learning

Data imbalance theory explains the negative impact of skewed class distributions on the performance of machine learning models. In intrusion detection datasets, normal traffic typically dominates, while attack instances are relatively scarce. This imbalance leads to biased learning,

where models tend to favor the majority class and perform poorly in detecting minority classes. The implications of this theory are significant in cybersecurity, as failure to detect minority attack classes can result in serious security breaches. Addressing data imbalance is therefore essential for improving model reliability and overall detection performance (Fernández et al., 2022).

2.3.4 Synthetic Data Generation Theory (SMOTE)

Synthetic data generation techniques are based on the principle of artificially increasing the representation of minority classes in a dataset. The Synthetic Minority Over-sampling Technique (SMOTE) operates by generating new data points through interpolation between existing minority class samples.

This approach is grounded in the assumption that expanding the feature space of minority classes improves the model’s ability to learn their underlying patterns. Compared to simple duplication methods, SMOTE reduces overfitting and enhances generalization. Its effectiveness in handling imbalanced datasets has been widely validated in intrusion detection studies (Fernández et al., 2022).

2.3.5 Summary of Theoretical Foundation

The theories discussed in this section collectively provide a strong foundation for the proposed system. Supervised learning enables effective classification, federated learning ensures privacy-preserving distributed training, data imbalance theory highlights the need for balanced datasets,

and SMOTE provides a practical solution for data generation. Together, these theories justify the integration of federated learning and SMOTE in improving intrusion detection performance.

2.4 Technical Foundations

This section presents the technical components and tools that form the basis for the design and implementation of the proposed federated learning-based intrusion detection system. It highlights the dataset used, pre-processing techniques, model design, federated learning framework, and data generation approach adopted in this study.

2.4.1 Dataset Description (NSL-KDD Dataset)

The NSL-KDD dataset is widely used as a benchmark for evaluating intrusion detection systems. It is an improved version of the KDD Cup 1999 dataset, designed to address issues such as redundant records and biased distributions. The dataset contains labeled network traffic data, where each instance represents a connection characterized by multiple features, including basic, content-based, and traffic-based attributes.

In this study, the NSL-KDD dataset is used for training and testing the proposed model. The dataset is divided into two subsets: the training set (KDDTrain+) and the testing set (KDDTest+). For simplicity and effective evaluation, the problem is formulated as a binary classification task, where network traffic is categorized as either normal or attack.

2.4.2 Data Preparation Techniques

Data preprocessing is a critical step in preparing the dataset for machine learning. The following preprocessing steps are applied in this study:

- i. **Data Cleaning:** Removal of unnecessary attributes such as the difficulty level column.
- ii. **Categorical Encoding:** Conversion of categorical features (e.g., protocol type, service, flag) into numerical representations using one-hot encoding.
- iii. **Feature Scaling:** Normalization of feature values using standard scaling to ensure uniform contribution of features during model training.
- iv. **Label Transformation:** Conversion of multi-class attack labels into binary classes (normal and attack) to simplify the classification problem.

2.4.3 Model Architecture

The proposed intrusion detection model is implemented using a feed-forward artificial neural network. The model consists of multiple fully connected layers with nonlinear activation functions, enabling it to learn complex patterns from network traffic data.

The architecture typically includes:

- i. Input layer corresponding to the number of features
- ii. Hidden layers with activation functions such as ReLU

iii. Output layer for classification

The model is trained using a cross-entropy loss function and optimized using gradient-based optimization techniques. This architecture is suitable for handling structured network data and provides a balance between performance and computational efficiency.

2.4.4 Federated Learning Techniques

The system adopts a federated learning framework in which the training dataset is distributed across multiple client nodes. Each client independently trains a local model using its own data and shares model updates with a central server.

The federated learning process involves the following steps: Initialization of a global model at the server

1. Distribution of the global model to clients
2. Local training at each client using its dataset
3. Transmission of model updates to the server
4. Aggregation of updates using Federated Averaging (FedAvg)
5. Update and redistribution of the global model

This iterative process continues for several rounds until the model converges. The approach ensures data privacy while enabling collaborative learning across distributed clients.

2.4.5 Data Generation Using SMOTE

To address class imbalance in the dataset, the Synthetic Minority Over-sampling Technique (SMOTE) is applied at the client level. SMOTE generates synthetic samples for minority classes by interpolating between existing instances, thereby improving class distribution.

In this study, SMOTE is integrated into the federated learning pipeline as a preprocessing step before local model training. Each client applies SMOTE to its local dataset, ensuring that minority classes are adequately represented during training. This approach enhances the model's ability to detect rare attack patterns and improves overall classification performance.

2.4.6 Performance Evaluation Metrics

The performance of the proposed system is evaluated using standard classification metrics, including:

- i. **Accuracy:** Measures the overall correctness of the model
- ii. **Precision:** Indicates the proportion of correctly predicted positive instances
- iii. **Recall:** Measures the ability of the model to detect actual positive instances
- iv. **F1-Score:** Provides a balance between precision and recall

In addition, confusion matrices and graphical plots are used to provide a detailed analysis of model performance. A comparative evaluation is

conducted between models trained with and without SMOTE to assess the effectiveness of the data generation approach.

2.5 Related Work

This section reviews recent studies related to intrusion detection systems, federated learning, and data generation techniques, with the aim of identifying existing approaches and highlighting gaps that justify the present study.

2.5.1 Federated Learning-Based Intrusion Detection

Recent research has demonstrated the growing relevance of federated learning in intrusion detection systems, particularly in addressing privacy concerns and distributed data challenges. Soomro et al. (2025) proposed a federated intrusion detection framework that leverages secure aggregation and decentralized training to enhance privacy while maintaining competitive detection performance. Their findings indicate that federated learning can effectively handle distributed data environments without compromising model accuracy.

Similarly, Nguyen et al. (2022) developed a federated deep learning-based IDS capable of detecting network intrusions across multiple nodes. Their study showed that federated models can achieve comparable performance to centralized models while significantly reducing the risk of data leakage. However, the authors noted that variations in client data distributions can affect model convergence.

In another study, Li et al. (2021) examined the impact of non-IID data in

federated learning systems and highlighted challenges related to model instability and reduced generalization. These findings are particularly relevant to intrusion detection, where data heterogeneity across network nodes is common.

2.5.2 Data Generation Techniques in Intrusion Detection

The issue of class imbalance in intrusion detection datasets has been widely studied, with various techniques proposed to improve model performance. Traditional methods such as SMOTE continue to be widely adopted due to their simplicity and effectiveness. Fernández et al. (2022) demonstrated that SMOTE significantly improves classification performance in imbalanced datasets by enhancing the representation of minority classes.

Recent studies have also explored advanced data generation techniques. For example, Albahar (2023) investigated the use of Generative Adversarial Networks (GANs) for intrusion detection and reported improved detection rates for minority attack classes. However, the study also highlighted challenges related to training complexity and computational cost. Although these advanced techniques show promise, their practical application in distributed systems remains limited due to resource constraints and implementation complexity.

2.5.3 Identified Research Gap

Based on the reviewed literature, the following gaps are identified:

- i. Many federated learning-based IDS models focus primarily on pri-

vacy preservation without adequately addressing data imbalance.

- ii. Existing data generation approaches within federated environments often rely on complex techniques such as GANs, which are computationally expensive and difficult to implement.
- iii. There is limited research on the application of simple and efficient data generation techniques, such as SMOTE, within federated learning frameworks.
- iv. Few studies provide practical implementation and evaluation using standard intrusion detection datasets in a simulated federated environment.

To address these gaps, this study proposes a federated learning-based intrusion detection system that integrates SMOTE as a data generation technique at the client level. The approach emphasizes simplicity, efficiency, and practical applicability while improving detection performance in distributed environments.

2.6 Chapter Summary

This chapter has examined the key concepts, theories, technical foundations, and existing studies related to intrusion detection systems, federated learning, and data generation techniques. It began by outlining the fundamental concepts underlying intrusion detection and distributed machine learning, followed by a clarification of key terms relevant to the study.

The analysis of theories provided insight into the principles of supervised machine learning, federated learning, data imbalance, and synthetic data generation, which collectively form the theoretical basis for the proposed system. These theories justify the adoption of a federated learning framework combined with SMOTE to improve intrusion detection performance in distributed environments.

Furthermore, the technical foundations section described the practical components required for system implementation, including the use of the NSL-KDD dataset, data preprocessing techniques, model architecture, federated learning framework, and performance evaluation metrics. This establishes a clear link between theoretical concepts and the actual system design.

The review of related works highlighted recent advancements in federated intrusion detection and data generation techniques. While existing studies demonstrate the potential of federated learning and advanced data generation methods, several limitations were identified, particularly in handling data imbalance using practical and computationally efficient approaches.

Based on these observations, this study addresses the identified gaps by proposing a federated learning-based intrusion detection system that integrates SMOTE as a lightweight and effective data generation technique. The next chapter presents the methodology adopted for the design and implementation of the proposed system.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter presents the methodology adopted for the design and implementation of the proposed federated learning-based intrusion detection system. It outlines the processes, techniques, and tools used to develop the model, as well as the procedures followed in data preparation, system design, and performance evaluation.

The study adopts an experimental approach in which a machine learning model is developed within a simulated federated learning environment. The system is designed to allow multiple clients to independently train local models using their respective datasets, while a central server aggregates the model updates to produce a global model. This approach ensures data privacy and supports distributed learning, which are essential in modern network environments.

Furthermore, the methodology incorporates a data generation technique, specifically the Synthetic Minority Over-sampling Technique (SMOTE), to address the issue of class imbalance commonly found in intrusion detection datasets. By applying SMOTE at the client level before model training, the system improves the representation of minority attack classes and enhances detection performance.

The subsequent sections of this chapter provide detailed descriptions of the system specifications, data collection process, model formulation,

system design, implementation procedures, and evaluation methods used in this study.

3.2 System Specification

This section outlines the technical requirements and specifications necessary for the design and implementation of the proposed federated learning-based intrusion detection system. The system specifications are categorized into hardware and software requirements to ensure proper deployment and execution of the model.

3.2.1 Hardware Requirements

The implementation of the proposed system does not require high-end computational resources, as the model is designed to operate efficiently within a simulated federated learning environment. The system was developed and tested on a standard personal computer with the following specifications:

1. Processor: Intel Core i5 or equivalent
2. Memory (RAM): Minimum of 8 GB
3. Storage: At least 256 GB available disk space
4. Operating System: Windows 10/11 (64-bit)

These specifications are sufficient for handling dataset preprocessing, model training, and evaluation tasks. The use of a CPU-based environment further demonstrates the practicality and accessibility of the proposed system.

3.2.2 Software Requirements

The system is implemented using widely adopted programming tools and machine learning libraries, ensuring flexibility and reproducibility. The key software components include:

1. Programming Language: Python (version 3.10 or higher)
2. Development Environment: Visual Studio Code (VS Code)
3. Libraries and Frameworks:
 - a) PyTorch: Used for building and training the neural network model
 - b) Pandas and NumPy: Used for data manipulation and numerical operations
 - c) Scikit-learn: Used for preprocessing, model evaluation, and performance metrics
 - d) Matplotlib: Used for data visualization and plotting results
 - e) Imbalanced-learn: Used for implementing SMOTE
4. Virtual Environment: Python virtual environment (venv) for dependency management

3.2.3 System Requirements and Constraints

The proposed system is designed with the following operational considerations:

- i. **Simulated Federated Environment:** The federated learning process is implemented on a single machine by partitioning the dataset into multiple client nodes.
- ii. **Data Dependency:** The system relies on the NSL-KDD dataset for training and evaluation.
- iii. **Binary Classification Setup:** The model simplifies intrusion detection into two classes: normal and attack.
- iv. **Conceptual Limitation:** The system avoids the use of computationally intensive models such as GANs to ensure efficiency.

3.3 Data Collection

This section describes the source and characteristics of the dataset used in this study, as well as the rationale for its selection.

3.3.1 Source of Data

The dataset used for this study is the **NSL-KDD dataset**, which is a widely recognized benchmark dataset for evaluating intrusion detection systems. It is an improved version of the KDD Cup 1999 dataset, developed to address issues such as redundant records and biased class distributions present in the original dataset.

The NSL-KDD dataset consists of labeled network traffic records, where each record represents a network connection characterized by a set of features describing various aspects of the traffic. The dataset includes both normal activities and different categories of cyber-attacks, making

it suitable for training and evaluating machine learning-based intrusion detection models.

The dataset is publicly available and has been extensively used in recent intrusion detection research, ensuring comparability and reliability of results (Tavallaei et al., 2009; recent usage supported in Ahmad et al., 2021).

3.3.2 Dataset Description

The NSL-KDD dataset contains 41 features describing network traffic behavior, along with a class label indicating whether the traffic is normal or an attack. These features can be grouped into:

- i. Basic Features: Attributes derived from TCP/IP connections (e.g., duration, protocol type)
- ii. Content Features: Information extracted from packet payloads (e.g., number of failed login attempts)
- iii. Traffic Features: Statistical measures computed over a time window (e.g., connection count)

The dataset is divided into two main subsets:

- a) KDDTrain+: Used for training the model
- b) KDDTest+: Used for testing and evaluation

3.3.3 Data Preparation for the Study

For the purpose of this study, the dataset is prepared to suit the requirements of the proposed model. The preparation process includes:

- i. **Binary Label Conversion:** All attack categories are grouped into a single class labeled *attack*, while normal traffic is labeled normal.
- ii. **Data Partitioning:** The training dataset is divided into multiple subsets to simulate distributed client nodes in a federated learning environment.
- iii. **Handling Data Imbalance:** Since attack instances are less frequent compared to normal traffic, SMOTE is applied at the client level to generate synthetic samples and improve class balance.

3.3.4 Justification for Data Collection

The NSL-KDD dataset is selected for this study due to the following reasons:

1. It is a standard benchmark dataset widely used in intrusion detection research
2. It provides labeled data suitable for supervised learning
3. It addresses some of the limitations of earlier datasets, such as redundancy and bias
4. It enables fair comparison with existing studies

Furthermore, its structured format and availability make it suitable for implementing and evaluating federated learning-based intrusion detection systems.

3.4 Model Formulation

This section presents the formulation of the proposed intrusion detection model, including the mathematical representation of the classification problem, the model structure, and the federated learning process used for training.

3.4.1 Problem Formulation

The intrusion detection task in this study is formulated as a binary classification problem, where the objective is to classify network traffic into two classes:

- i. Class 0: Normal traffic
- ii. Class 1: Attack traffic

Let the Dataset be represented as:

$$D = \{(x_i, y_i)\}_{i=1}^N$$

Where:

- 1. $X_i \in \mathbb{R}^n$ represents the feature vector of the iii -th network instance.
- 2. $y_i \in \{0, 1\}$ represents the corresponding class label.
- 3. N is the total number of samples

The goal of the model is to learn a function:

$$f(x; \theta) \rightarrow y$$

where θ represents the model parameters, such that the predicted output $\hat{y}$ closely matches the true label y

3.4.2 Model Structure

The proposed model is implemented using a feedforward artificial neural network (ANN). The model consists of:

- i. An input layer corresponding to the number of features
- ii. One or more hidden layers with nonlinear activation functions (ReLU)
- iii. An output layer with two neurons representing the binary classes

The model outputs class probabilities, and the predicted class is obtained using the argmax function.

The model is trained using the cross-entropy loss function, defined as:

$$L = - \sum_{i=1}^N y_i \log(\hat{y}_i)$$

Where:

y_i is the true label

$\hat{y}_i$ is predicted Probability

This loss function is suitable for classification tasks and helps in minimizing prediction error during training.

3.4.3 Federated Learning Formulation

In the federated learning setting, the dataset is distributed across multiple clients. Each client k has its local dataset D_k and trains a local model with parameters θ_k .

The objective is to minimize the global loss function:

$$F(\theta) = \sum_{k=1}^K \frac{n_k}{n} F_k(\theta)$$

Where:

K is the number of client

n_k is the number of samples at client k

n is the total number of samples

$F_k(\theta)$ is the local loss function at client k

Each client performs local training and sends its updated model parameters to the central server.

3.4.4 Model Aggregation (Federated Averaging)

The global model is updated using the Federated Averaging (FedAvg) algorithm, which aggregates local model parameters as follows: This

$$\theta = \sum_{k=1}^K \frac{n_k}{n} \theta_k$$

weighted averaging ensures that clients with larger datasets have a greater influence on the global model.

3.4.5 Integrated of SMOTE in Model Training

To address class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) is applied at each client before local model training. Let the original dataset at a client be D_k then SMOTE generates an augmented dataset: Where $D_{\text{synthetic}}$ represents the generated minority class

$$D'_k = D_k \cup D_{\text{synthetic}}$$

samples.

By balancing the class distribution at each client, SMOTE improves the learning process and enhances the model's ability to detect minority attack classes.

3.4.6 Training Objective

The overall objective of the system is to minimize the global loss function while improving classification performance across all clients. This is achieved through:

- a. Local model training using balanced datasets
- b. Iterative aggregation of model parameters
- c. Continuous updating of the global model

This formulation ensures that the proposed system effectively combines federated learning and data generation techniques to improve intrusion detection performance.

3.5 System Design

This section describes the overall design of the proposed federated learning-based intrusion detection system. It presents the system architecture, workflow, and the interaction between different components involved in data processing, model training, and evaluation.

3.5.1 Overview of the Proposed System

The proposed system is designed as a distributed intrusion detection framework based on federated learning. The system consists of multiple client nodes and a central server that collaboratively train a global model without sharing raw data.

Each client independently performs data preprocessing, applies SMOTE to address class imbalance, and trains a local model. The trained local models are then sent to the central server, where they are aggregated to produce a global model. This process is repeated over multiple communication rounds to improve model performance.

3.5.2 System Architecture

The architecture of the proposed system is composed of the following key components:

- i. **Client Nodes:** Responsible for local data storage, preprocessing, SMOTE application, and model training.
- ii. **Central Server:** Aggregates model updates received from clients and maintains the global model.

- iii. **Communication Layer:** Facilitates the exchange of model parameters between clients and the server.
- iv. **Global Model:** The final model obtained after aggregating local models from all clients.

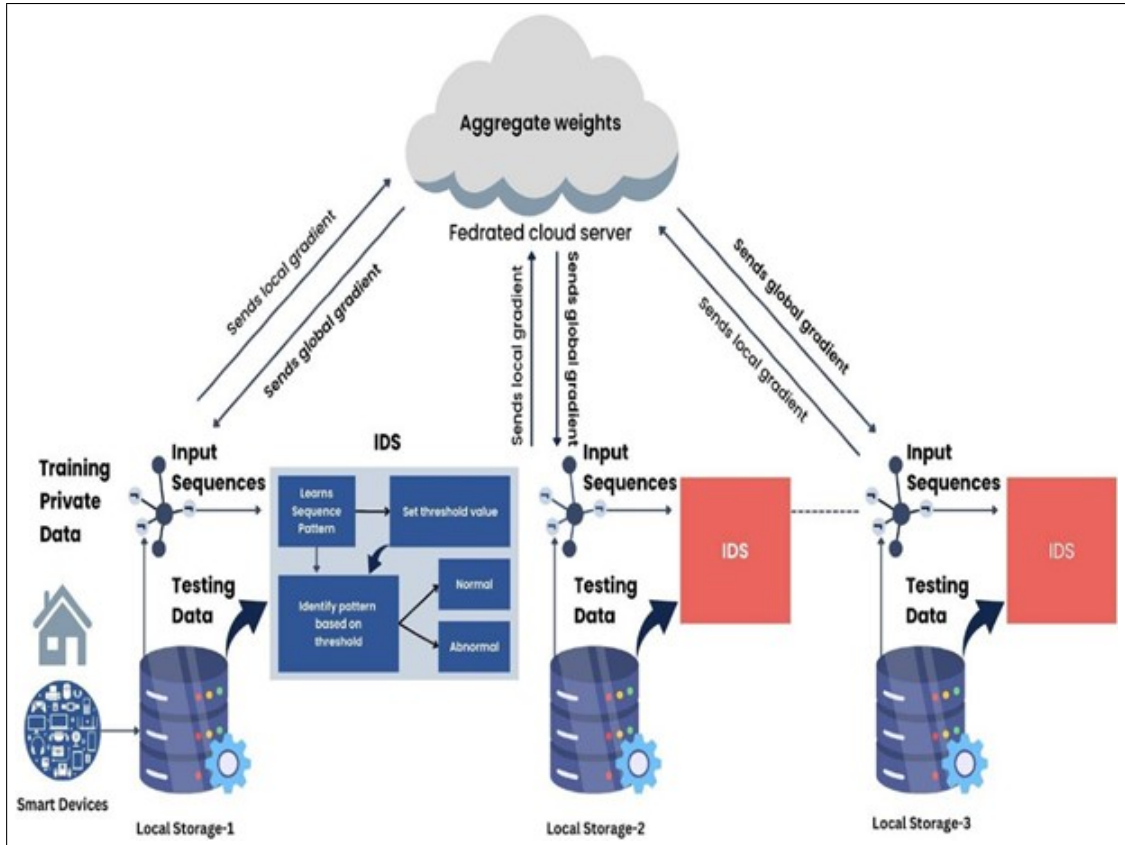


Figure 3.1: Proposed Federated Intrusion Detection System Architecture

3.5.3 System Workflow

The operation of the proposed system follows a structured workflow:

- i. **Data Loading:** The NSL-KDD dataset is loaded and partitioned into multiple subsets representing different clients.
- ii. **Data Preprocessing:** Each client preprocesses its local dataset through encoding, normalization, and label transformation.

- iii. **SMOTE Application:** SMOTE is applied locally to balance class distribution within each client dataset.
- iv. **Local Model Training:** Each client trains a local neural network model using its processed dataset.
- v. **Model Aggregation:** Local model parameters are sent to the central server and aggregated using the FedAvg algorithm.
- vi. **Global Model Update:** The updated global model is redistributed to clients for further training.
- vii. **Model Evaluation:** The final global model is evaluated using test data and performance metrics.

3.5.4 System Design Flow

The logical flow of the system can be represented as a step-by-step process from data input to output prediction.

3.5.5 Design Consideration

The design of the proposed system is guided by the following considerations:

- i. **Data Privacy:** Ensuring that raw data remains at the client level.
- ii. **Scalability:** Allowing the system to support multiple clients.
- iii. **Efficiency:** Using lightweight techniques such as SMOTE instead of computationally expensive models.

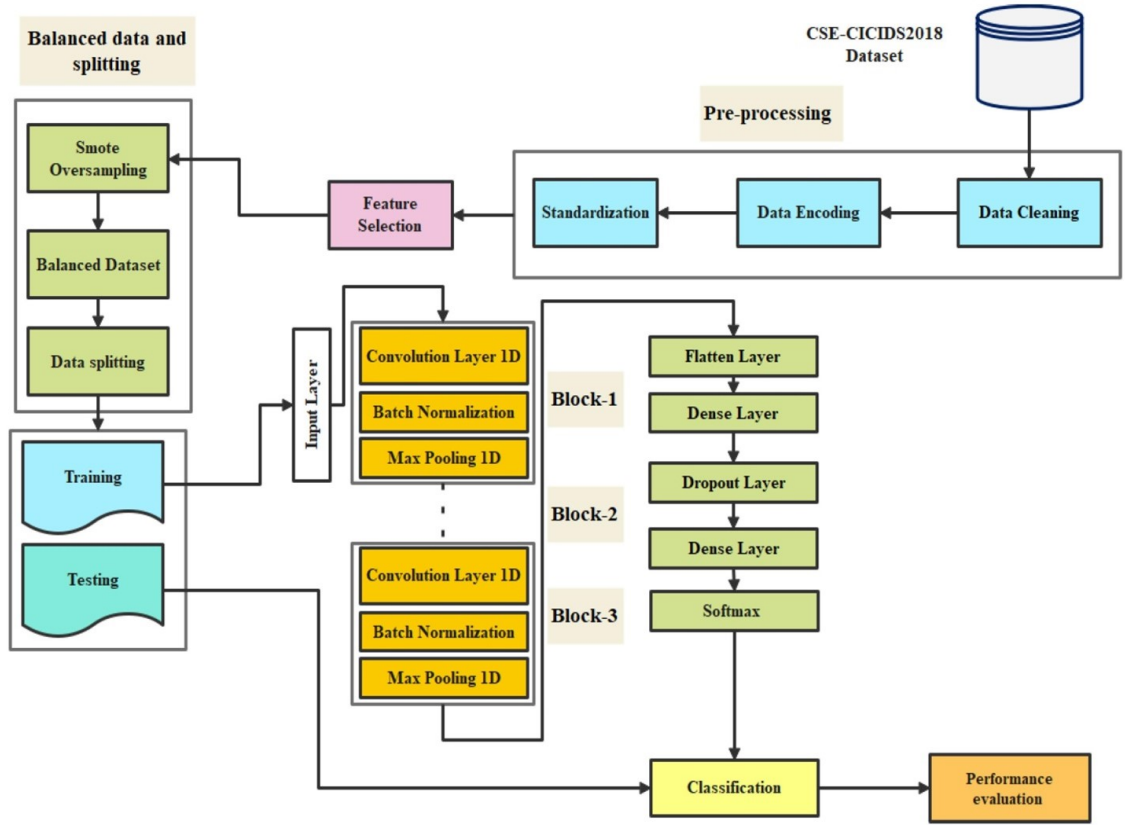


Figure 3.2: System Workflow of the Proposed Model

- iv. **Performance Optimization:** Improving detection accuracy, especially for minority classes.

3.6 System Implementation

This section describes the practical implementation of the proposed federated learning-based intrusion detection system. It outlines how the system components were developed, integrated, and executed using the selected tools and technologies.

3.6.1 Development Environment

The system was implemented using the Python programming language due to its flexibility and strong support for machine learning applications. The development was carried out using Visual Studio Code (VS Code)

as the integrated development environment.

A virtual environment was created to manage project dependencies, ensuring proper isolation of required libraries. Key libraries used in the implementation include:

- a. **PyTorch:** for building and training the neural network model
- b. **Pandas and NumPy:** for data manipulation and numerical operations
- c. **Scikit-learn:** for preprocessing and evaluation metrics
- d. **Matplotlib:** for visualization of results
- e. **Imbalanced-learn:** for implementing SMOTE

3.6.2 Federated Learning Implementation

The federated learning process was simulated on a single machine by partitioning the training dataset into multiple subsets, each representing a client node. Each client independently trains a local model using its dataset.

The Implementation follows these steps:

- i. Initialization of a global model
- ii. Distribution of the global model to all clients
- iii. Local training of models at each client
- iv. Collection of model parameters from clients

- v. Aggregation of parameters using the Federated Averaging (FedAvg) algorithm
- vi. Updating and redistribution of the global model

3.6.3 Integration of SMOTE

To address the issue of class imbalance, SMOTE was applied at each client before local model training. The imbalanced-learn library was used to generate synthetic samples for the minority class.

By applying SMOTE locally, each client dataset becomes more balanced, allowing the model to learn better representations of both normal and attack classes. This integration plays a key role in improving the overall detection performance of the system.

3.6.4 Implementation of the Result Storage

- i. Performance metrics (accuracy, precision, recall, F1-score)
- ii. Confusion Matrices
- iii. Classification reports
- iv. Loss plots
- v. Comparison results (with SMOTE vs without SMOTE)

These outputs were saved in formats such as CSV, JSON, and image files, enabling easy integration into Chapter 4 of this study.

3.7 System Evaluation

This section describes the methods and criteria used to evaluate the performance of the proposed federated learning-based intrusion detection system. The evaluation focuses on measuring the effectiveness of the model in correctly classifying network traffic and assessing the impact of the SMOTE-based data generation technique.

3.7.1 Evaluation Approach

The evaluation of the proposed system is carried out using a supervised learning evaluation approach, where the trained global model is tested on unseen data from the NSL-KDD test dataset. The model's predictions are compared with the actual labels to determine its classification performance.

To assess the effectiveness of the data generation technique, two experimental scenarios are considered:

- i. **Scenario 1:** Federated learning without SMOTE
- ii. **Scenario 2:** Federated learning with SMOTE

The results obtained from both scenarios are compared to evaluate the contribution of SMOTE to the overall system performance.

3.7.2 Performance Metrics

The performance of the model is evaluated using standard classification metrics commonly used in intrusion detection research:

- i. Accuracy: Measures the overall proportion of correctly classified instances.
- ii. Precision: Indicates the proportion of correctly predicted attack instances among all predicted attack instances.
- iii. Recall (Detection Rate): Measures the ability of the model to correctly identify actual attack instances.
- iv. F1-Score: Provides a balance between precision and recall, especially useful for imbalanced datasets.

3.7.3 Confusion Matrix Analysis

A confusion matrix is used to provide a detailed breakdown of the model's predictions. It shows the number of:

- i. True Positives (correctly detected attacks)
- ii. True Negatives (correctly identified normal traffic)
- iii. False Positives (normal traffic incorrectly classified as attack)
- iv. False Negatives (attacks incorrectly classified as normal)

3.7.4 Comparative Evaluation

To demonstrate the effectiveness of the proposed approach, a comparative evaluation is conducted between the models trained with and without SMOTE. The comparison focuses on:

- i. Improvement in detection of minority classes
- ii. Changes in precision, recall, and F1-score

- iii. Overall model accuracy

3.7.5 Visualization of Result

Graphical tools are used to enhance the interpretation of results. These include:

- i. Confusion matrix plots
- ii. Loss curves showing training progress
- iii. Bar charts comparing performance metrics

These visualizations provide an intuitive understanding of the model's behavior and performance trends.

3.8 Chapter Summary

This chapter has presented the methodology adopted for the design and implementation of the proposed federated learning-based intrusion detection system. It began by outlining the system specifications, including the hardware and software requirements necessary for the development of the model.

The chapter further described the data collection process, highlighting the use of the NSL-KDD dataset as a benchmark for intrusion detection. The dataset was preprocessed through feature encoding, normalization, and label transformation to ensure suitability for machine learning tasks. The model formulation section defined the classification problem, the neural network structure, and the mathematical basis of the federated

learning approach, including the use of the Federated Averaging algorithm.

In addition, the system design section presented the architecture and workflow of the proposed system, illustrating how multiple clients collaboratively train a global model while maintaining data privacy. The implementation section detailed the practical development of the system using Python and relevant machine learning libraries, including the integration of SMOTE as a data generation technique to address class imbalance.

Finally, the system evaluation section described the performance metrics and evaluation procedures used to assess the effectiveness of the model, including a comparative analysis between models trained with and without SMOTE.

Overall, this chapter provides a comprehensive description of the processes and techniques used to develop the proposed system. The next chapter presents the results obtained from the implementation and discusses the performance of the model based on the evaluation metrics.

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